

Neural Network Project - Engineering Report

1. Linear Algebra Core (Matrix class)

The system is powered by a custom Matrix engine. - **Efficiency:** Implements Move Semantics ($O(1)$) and deep copy logic. - **Math Operations:** Optimized multiplication, Hadamard products, and ReLU derivatives.

2. Neural Architecture (Layer class)

Each layer manages weights and biases for signal processing. - **Initialization:** Xavier/He initialization ($1/\sqrt{\cdot}$) for stability. - **Activation:** Integrated ReLU processing.

3. High-Level Orchestration (Network class)

The top-level orchestrator for the AI model.